Yash Vardhan Pant

Contact Information

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Waterloo, ON, N2L 0J8, Canada

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Research Interests

Control Theory, Formal Methods, Machine Learning and Optimization with applications in Control and Motion Planning for Autonomous Mobile Robots and other Cyber-Physical Systems (CPS).

Experience

Assistant Professor July 2021 – Present
Department of Electrical and Computer Engineering (ECE),
University of Waterloo, Waterloo, ON, Canada
Courtesy Appointment, David R. Cheriton School of Computer Science.
Extended Research Team, Waterloo RoboHub.
Faculty Affiliate, Waterloo AI institute.
Faculty Affiliate, Computational Mathematics.
Faculty Affiliate, Waterloo Institute of Sustainable Aeronautics (WISA)
Faculty Affiliate, Waterloo Cybersecurity and Privacy Institute.

Postdoctoral Fellow: NSF VeHICaL Project October 2019 – July 2021
Department of Electrical Engineering and Computer Sciences,
University of California, Berkeley, CA
Mentors: Prof. Sanjit Seshia, Prof. Bjoern Hartmann)

Research Intern, Control Systems May 2014 – Aug 2014
Lit Motors, San Francisco, CA
Supervisor: Dr. Berenice Mettler

Education

Doctor of Philosophy (Ph.D.) in Electrical Engineering Sep 2012 – Aug 2019
University of Pennsylvania (U. Penn)
Thesis Title: Robust Predictive Methods for Planning and Control of Autonomous Systems
Committee: Profs. George Pappas (Chair), Manfred Morari, Georgios Fainekos & Jyo Deshmukh
Advisor: Prof. Rahul Mangharam

Master of Science (M.S.) in Electrical Engineering Sep 2010 – May 2012
University of Pennsylvania
Thesis Title: AutoPlug: A Testbed for Automotive Control Software/Hardware Diagnostics and Remote Recalls Management
Advisor: Prof. Rahul Mangharam

Bachelor of Technology (B.Tech) in Electronics & Telecom. Engineering Aug 2006 – Jun 2010
College of Engineering Roorkee, India

Honors and Awards

Outstanding Performance Award (Engineering), University of Waterloo. Waterloo, Canada, 2025
Research Excellence Award: ECE, University of Waterloo. Waterloo, Canada, 2025
Best Paper Award: Canadian Conference on Artificial Intelligence (CAIAC). Guelph, Canada, 2024
Best Paper in Session: AIAA/IEEE Digital Avionics Systems Conference (DASC) USA, 2020
Best Student Paper Award: IEEE NAECON Dayton, Ohio, USA 2018
Student Travel Grant: IEEE CCTA Big Island, Hawaii, USA 2017
Student Travel Grant: ACC Portland, USA 2014
Richard K. Dentel Memorial Prize for outstanding research in Urban Transportation U. Penn 2013

Top-10 finish, Intel Cornell Cup competition — Orlando, USA 2013
Third place, World Embedded Software Competition — Seoul, South Korea 2013
Student Travel Grant: ACM HiCoNS (CPS Week) — Beijing, China 2012
Second place, The Embedded System Competition — Indian Institute of Technology Roorkee, India 2009

Journal Publications

*Author-order key: authors are listed by contribution, with the supervising author(s) last. Co-authors supervised by Y. V. Pant are marked * (graduate student) or ** (postdoctoral researcher / research associate); † denotes equal (co-first) authorship.*

[1] J. Zhong*, L. Khoshnevisan, S. Huang, M. Pirani**, **Y. V. Pant**, A. Khajepour. Multi-Objective Agent-Based Model Predictive Controller for Plug-and-Play Vehicle Control. *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 12, pp. 22818–22829, Dec. 2025. (11 pages, double column)

[2] Z. F. Dai*, **Y. V. Pant**, S. L. Smith. Model Predictive Control for Systems With Partially Unknown Dynamics Under Signal Temporal Logic Specifications. *IEEE Control Systems Letters*, vol. 8, pp. 2931–2936, Dec. 2024. (6 pages, double column)

[3] J. Zhong*, R. Valiollahi Mehrizi, M. Pirani**, Y. Chao, A. Kasaiezadeh, **Y. V. Pant**, A. Khajepour. Learning Agent-based Model Predictive Control for Holistic Vehicle Performance. *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 17482–17492, Nov. 2024. (11 pages, double column)

[4] J. Wang**, **Y. V. Pant**, L. Zhao, M. Antkiewicz, K. Czarnecki. Enhancing Safety in Mixed Traffic: Learning-Based Modeling and Efficient Control of Autonomous and Human-Driven Vehicles. *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 12784–12799, Sep. 2024. (16 pages, double column)

[5] J. Wang**, Z. Jiang, **Y. V. Pant**. Learning-Based Modeling of Human-Autonomous Vehicle Interaction for Improved Safety in Mixed-Vehicle Platooning Control. *Transportation Research: Part C* (Elsevier), vol. 162, Art. no. 104600, May 2024. (13 pages, single column)

[6] J. Wang**, Z. Jiang, **Y. V. Pant**. Improving Safety in Mixed Traffic: A Learning-based Model Predictive Control for Autonomous and Human-Driven Vehicle Platooning *Knowledge-Based Systems* (Elsevier), vol. 293, Art. no. 111673, Jun. 2024. (12 pages, double column)

[7] **Y. V. Pant** , M. Z. Li, R. A. Quaye, A. Rodionova, H. Abbas, M. Ryerson, R. Mangharam. FADS: Framework for Autonomous Drone Safety, *Transportation Research Part C: Emerging Technologies* (Elsevier), vol. 130, Art. no. 103275, Sep. 2021, special issue on Embracing Urban Air Mobility. (22 pages, single column)

[8] A. Rodionova, **Y. V. Pant**, C. Kurtz, K. J. Jang, H. Abbas, R. Mangharam. Learning-'N-Flying: A Learning-based, Decentralized Mission Aware UAS Collision Avoidance Scheme. *ACM Transactions on Cyber-Physical Systems*, vol. 5, no. 4, Art. no. 35, Oct. 2021. (26 pages, single column)

[9] **Y. V. Pant**, H. Abbas, K. Mohta, R. A. Quaye, T. X. Nghiem, J. Devietti, R. Mangharam. Anytime Computation and Control for Autonomous Systems. *IEEE Transactions on Control Systems Technology*, vol. 29, no. 2, pp. 768–779, 2020. (12 pages, double column)

Conference Publications

[1] D. S. Lee*, **Y. V. Pant**, S. Fischmeister. Top-down Traffic Scenario Generation via Joint Initial-Goal Diffusion and Trajectory Infilling *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Naples, Italy, Sep. 2026. (7 pages, double column)

[2] K. Kalavadia*, S. Dutta**, **Y. V. Pant**, S. L. Smith. Efficient Multi-Objective Planning with Weighted Maximization using Large Neighbourhood Search. *IEEE International Conference on Robotics*

and Automation (ICRA), Vienna, Austria, Jun. 2026. (8 pages, double column)

[3] A. Iskandar[*], S. Dutta[**], K. Murrant, **Y. V. Pant**, S. L. Smith. Hierarchical Informative Path Planning via Graph Guidance and Trajectory Optimization. *American Control Conference (ACC)*, New Orleans, USA, May 2026 (8 pages, double column)

[4] A. Karthikeyan[*], **Y. V. Pant**. GenPlan: Generative Sequence Models as Adaptive Planners. *AAAI Conference on Artificial Intelligence*, Philadelphia, USA, Feb. 2025. (8 pages, single column; 6-page appendix)

[5] P. Kolapalli[*], **Y. V. Pant**, S. L. Smith. Drive-by-Logic: Trajectory Generation for Nonholonomic Ground Robots with Signal Temporal Logic Objectives. *IEEE Conference on Control Technology and Applications (CCTA)*, San Diego, USA, Aug. 2025. (8 pages, double column)

[6] D. S. Lee[*†], A. Karthikeyan[*†], **Y. V. Pant**, S. Fischmeister. Path Diffuser: Diffusion Model for Data-Driven Traffic Simulator. *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Gold Coast, Australia, Nov. 2025. (8 pages, double column)

[7] N. Fronda, P. Smith, B. Hoxha, **Y. V. Pant**, H. Abbas. Fair-CoPlan: Negotiated Flight Planning with Fair Deconfliction for Urban Air Mobility. *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Gold Coast, Australia, Nov. 2025. (6 pages, double column)

[8] L. D'Souza[*], **Y. V. Pant**, S. Fischmeister. Tractable Stochastic Hybrid Model Predictive Control Using Gaussian Processes for Repetitive Tasks in Unseen Environments. *European Control Conference (ECC)*, Thessaloniki, Greece, Jun. 2025. (8 pages, double column)

[9] A. Alipanah[*], **Y. V. Pant**. Generating Malicious Demonstration Policies to Exploit Vulnerabilities in Inverse Reinforcement Learning. *Canadian AI Conference*, Calgary, Canada, May 2025. (12 pages, single column)

[10] A. Karthikeyan[*], **Y. V. Pant**. Adaptformer: Sequence Models as Adaptive Iterative Planners. *Conference on Robots and Vision*, Calgary, Canada, May 2025. (8 pages, double column)

[11] J. Zhong[*], R. Valiollahi Mehrizi, **Y. V. Pant**, A. Khajepour. A Data-Driven Distributed Control Scheme: Learning Multi-Objective Agent-Based MPC for Path-Tracking. *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, Canada, Sep. 2024. (6 pages, double column)

[12] S. M. Moghaddam[*], **Y. V. Pant**, S. Fischmeister. Robust Reinforcement Learning for Linear Temporal Logic Specifications with Finite Trajectory Duration. *Canadian AI Conference*, Guelph, Canada, May 2024. (12 pages, single column) **Best Paper Award**

[13] L. D'Souza[*], **Y. V. Pant**, Stochastic Hybrid Model Predictive Control using Gaussian Processes for Systems with Piecewise Residual Dynamics. *American Control Conference (ACC)*, San Diego, USA, May 2023. (8 pages, double column)

[14] **Y. V. Pant**, H. Abbas, R. Mangharam, Distributed Trajectory Planning for Multi-Rotor UAVs with Signal Temporal Logic Objectives. *IEEE Conference on Control Technology and Applications (CCTA)*, Trieste, Italy, Aug. 2022. (8 pages, double column)

[15] **Y. V. Pant**, B. T. Kumaravel, A. Shah, E. Kraemer, M. Vazquez-Chanlatte, K. Kulkarni, B. Hartmann, S. Seshia, Modeling and Influencing Human Attentiveness in Autonomy-To-Human Perception Hand-offs. *IEEE Conference on Intelligent Transportation Systems (ITSC)*, Macau, China, Oct. 2022. (8 pages, double column)

[16] C. Chin, M. Z. Li, **Y. V. Pant**, Distributed Traffic Flow Management for Uncrewed Aircraft Systems. *IEEE Conference on Intelligent Transportation Systems (ITSC)*, Macau, China, Oct. 2022. (7 pages, double column)

[17] M. Pirani[**], Y. She, R. Tang, Z. Jiang, **Y. V. Pant**. Stable Interaction of Autonomous Vehicle Platoons with Human-Driven Vehicles, *American Control Conference (ACC)*, Atlanta, USA, Jun. 2022. (8 pages, double column)

[18] **Y. V. Pant**[†], H. Yin[†], M. Arcak, S. A. Seshia, Co-design of Planning and Control for multi-rotor UAVs with Temporal Logic Objectives, *American Control Conference (ACC)*, New Orleans, USA, May 2021. (8 pages, double column)

[19] K. Viswanadha, F. Indaheng, J. Wong, E. Kim, E. Kalvan, **Y.V. Pant**, D. J. Fremont, S. A. Seshia. Addressing the IEEE AV test challenge with Scenic and VerifAI, *IEEE International Conference on Artificial Intelligence Testing (AITest)*, Oxford, UK, Aug. 2021. (7 pages, double column)

[20] S. Ghosh, **Y. V. Pant**, H. Ravanbakhsh, S. A. Seshia, Counterexample-Guided Synthesis of Perception Models and Control, *American Control Conference (ACC)*, New Orleans, USA, May 2021. (8 pages, double column)

[21] V. Tuck, **Y. V. Pant**, S. A. Seshia, S. S. Sastry. DEC-LOS-RRT: Decentralized path planning for multi-robot systems with Line-of-sight constrained communication. *IEEE Conference on Control Technology and Applications (CCTA)*, San Diego, USA, Aug. 2021. (8 pages, double column)

[22] A. Rodionova[†], **Y. V. Pant**[†], K. J. Jang, H. Abbas, R. Mangharam. Learning-to-Fly: Learning-based Collision Avoidance for Scalable Urban Air Mobility. *IEEE Conference on Intelligent Transportation Systems (ITSC)*, Rhodes, Greece, Sep. 2020. (8 pages, double column)

[23] K. J. Jang, **Y. V. Pant**, A. Rodionova, H. Abbas, R. Mangharam. Learning-to-Fly Faster: Reinforcement Learning-based UAV Collision Avoidance. *AIAA/IEEE Digital Avionics Systems Conference (DASC)*, Oct. 2020. (10 pages, double column) **Best Paper in Session Award**

[24] D. J. Fremont, E. Kim, **Y. V. Pant**, S. A. Seshia, A. Acharya, X. Bruso, P. Wells, S. Lemke, Q. Lu, S. Mehta. Formal Scenario-Based Testing of Autonomous Vehicles: From Simulation to the Real World. *IEEE Conference on Intelligent Transportation Systems (ITSC)*, Rhodes, Greece, Sep. 2020. (8 pages, double column)

[25] **Y. V. Pant**, R. A. Quaye, H. Abbas, A. Varre, R. Mangharam. Fly-by-Logic: A Tool for Unmanned Aircraft System Fleet Planning using Temporal Logic. *NASA Formal Methods Symposium*, Houston, USA, May 2019. (7 pages, single column)

[26] K. J. Jang, **Y. V. Pant**, B. Zhang, J. Weimer and R. Mangharam. Robustness Evaluation of Computer-aided Clinical trials for Medical Devices. *ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, Montreal, Canada, Apr. 2019. (11 pages, double column)

[27] H. Abbas, **Y. V. Pant**, R. Mangharam. Temporal Logic Robustness for General Signal Classes. *ACM International Conference on Hybrid Systems: Computation and Control (HSCC)*, Montreal, Canada, Apr. 2019. (12 pages, double column)

[28] **Y. V. Pant**, H. Abbas, R. A. Quaye, R. Mangharam. Fly-by-Logic: Control of Multi-Drone Fleets with Temporal Logic Objectives. *ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, Porto, Portugal, Apr. 2018. (12 pages, double column)

[29] M. Z. Li, W. R. Tan, S. M. Prakash, J. F. Kearney, M. S. Ryerson, D. Lee, **Y. V. Pant**. Design and implementation of a centralized system for autonomous unmanned aerial vehicle trajectory conflict resolution. *IEEE National Aerospace and Electronics Conference (NAECON)*, Dayton, USA, Jul. 2018. (9 pages, double column) **Best Student Paper Award**.

[30] **Y. V. Pant**[†], H. Abbas[†], R. Mangharam. Smooth Operator: Control of systems using the Smooth Robustness of Temporal Logic. *IEEE Conference on Control Technology and Applications (CCTA)*, Kohala Coast, USA, Aug. 2017. (6 pages, double column) **IEEE CCTA Student Travel Award** .

[31] **Y. V. Pant**, H. Abbas, R. Mangharam. Robust Model Predictive Control for Non-Linear Systems with Input and State Constraints via Feedback Linearization. *IEEE Conference on Decision and Control (CDC)*, Las Vegas, USA, Dec. 2016. (6 pages, double column)

[32] **Y. V. Pant**, H. Abbas, K. Mohta, T. X. Nghiem, J. Devietti, R. Mangharam. Co-design of Anytime Computation and Robust Control. *IEEE Real-Time Systems Symposium (RTSS)*, San Antonio, USA, Dec. 2015. (10 pages, double column)

[33] **Y. V. Pant**, H. Abbas, K. N. Nischal, P. Kelkar, D. Kumar, J. Devietti, R. Mangharam. Power-efficient algorithms for autonomous navigation. *IEEE Conference on Complex Systems Engineering (ICCSE)*, Storrs, USA, Nov. 2015. (6 pages, double column)

[34] **Y. V. Pant**, T. X. Nghiem, R. Mangharam .Peak power reduction in hybrid energy systems with limited load forecasts. *American Control Conference (ACC)*, Portland, USA, Jun. 2014. (6 pages, double column) **ACC Student Travel Award**

[35] U. Drolia[†], Z. Wang[†], **Y. V. Pant**[†], R. Mangharam. Autoplug: An automotive test-bed for electronic controller unit testing and verification. *IEEE Intelligent Transportation Systems Conference (ITSC)*, Washington, DC, USA, Oct. 2011. (6 pages, double column)

Workshop papers, Demo Abstracts and Works-in-Progress

[1] A. Karthikeyan[*], **Y. V. Pant**. DiffFP: Learning Behaviors from Scratch via Diffusion-based Fictitious-Play, IJCAI 2025 Spring Symposium on User-Aligned Assessment of Adaptive AI Systems, 2025.

[2] L. D'Souza[*], A. Karthikeyan[*], **Y. V. Pant**, S. Fischmeister. Learning Adaptive Diffusion Policies for Hybrid Dynamical Systems. IJCAI 2025 Spring Symposium on User-Aligned Assessment of Adaptive AI Systems, 2025.

[3] A. Alipanah[*], **Y. V. Pant**. Exploiting Demonstration Vulnerabilities: Malicious Agents in Inverse Reinforcement Learning. Cybersecurity and Privacy Institute (CPI) Graduate Student Conference, 2025.

[4] L. D'Souza[*], **Y. V. Pant**, S. Fischmeister. Learning-Based Hybrid Model Predictive Control for Systems With Partially Known and Evolving Dynamics. AAAI 2024 Spring Symposium on User-Aligned Assessment of Adaptive AI Systems, 2024.

[5] A. Karthikeyan[*], **Y. V. Pant**. AdaptFormer: Sequence Models as Adaptive Iterative Planners. AAAI 2024 Spring Symposium on User-Aligned Assessment of Adaptive AI Systems, 2024.

[6] D. J. Fremont, E. Kim, **Y. V. Pant**, S. A. Seshia, A. Acharya, X. Bruso, P. Wells, S. Lemke, Q. Lu, S. Mehta. Poster: Formal Scenario-Based Testing of Autonomous Vehicles. Automated Vehicles Symposium (AVS), 2020.

[7] **Y. V. Pant**, H. Abbas, R. Mangharam. Distributed planning of Multi-rotor drone fleets using the Smooth Robustness of Signal Temporal Logic. *Monitoring and Testing of CPS Workshop (MTCPS), CPS Week*, 2019.

[8] **Y. V. Pant**, H. Abbas, R. Mangharam. Control with Temporal Logic Requirements (poster). *SRC TECHCON*, 2017.

[9] **Y. V. Pant**, H. Abbas, R. Mangharam. Control using the Smooth Robustness of Temporal Logic. *Monitoring and Testing of CPS Workshop (MTCPS), CPS Week*, 2017.

[10] K. N. Nischal, P. Kelkar, D. Kumar, **Y. V. Pant**, H. Abbas, J. Devietti, R. Mangharam. Hardware Optimizations for Anytime Perception and Control. *Work-in-progress, Real-Time Systems Symposium (RTSS)*, 2015.

[11] P. Gurniak, **Y. V. Pant**. Demo: Low-cost Autonomous Navigation with Anytime Control and

Computation. *University Transportation Center (UPenn-CMU) Annual Meeting*, 2014.

[12] **Y. V. Pant**, T. X. Nghiem, R. Mangharam. Knock NOx: Model-based Remote Diagnostics of a Diesel Exhaust Control System. *Work-in-progress, IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*, 2013.

[13] **Y. V. Pant**, H. Jain, A. Mulay, R. Dutta. Protodrive: Rapid Prototyping and Simulation of Electric Vehicle Powertrains. *Final report: Intel Cornell Cup*, 2013. **Award for Top-10 Finish**

[14] **Y. V. Pant**, S. Diaz, H. Jain, W. Price, A. Botelho. Protodrive: Simulation of Electric Vehicle Powertrains. *Final report: World Embedded Software Competition*, 2013. **Third Place Award**

[15] W. Price, H. Jain, **Y. V. Pant**, R. Mangharam. Protodrive: An experimental platform for electric vehicle energy scheduling and control. *Work-in-progress, Real-Time Systems Symposium (RTSS)*, 2012.

[16] S. Diaz, H. Jain, **Y. V. Pant**, W. Price, R. Mangharam. Protodrive: An experimental platform for electric vehicle energy scheduling and control. *Demo Abstract Real-Time Systems Symposium (RTSS)*, 2012.

[17] **Y. V. Pant**. Demo: AutoPlug 2.0. *Real-Time and Embedded Technology and Applications Symposium (RTAS): Demo session*, 2012.

[18] **Y. V. Pant**, R. Mangharam. Observer-based Sensor Fault Detection and Isolation. *Work-in-progress, ACM International Conference on High Confidence Networked Systems (HiCoNS)*, 2012. **HiCoNS Student Travel Award**

Patent

United States Patent No. 11,429,118 for CONTROL OF MULTI-DRONE FLEETS WITH TEMPORAL LOGIC OBJECTIVES — Granted, August 2022

Research Grants

All amounts in Canadian dollars (CAD). "Applicant's share" is the percentage of the total award allocated to the candidate (100% for sole-PI grants and contracts). Granting agencies: NSERC — Natural Sciences and Engineering Research Council of Canada; NRC — National Research Council of Canada; NCC — National Cybersecurity Consortium; CFI — Canada Foundation for Innovation (JELF — John R. Evans Leaders Fund); ORF — Ontario Research Fund; WISA — Waterloo Institute for Sustainable Aeronautics; CPI — Cybersecurity and Privacy Institute.

Data-driven Digital Twins for Rotorcraft Dynamics — 2026 – 2027
- Research consulting grant ($38,500 CAD; applicant's share: 100%)
- Research Partner: NRC.

Holistic Robot Farming — 2026 – 2031
- NSERC Alliance Grant ($3,333,000 CAD; applicant's share: 16.7%)
- Industry Partners: Haggerty AgRobotics and BHF Agrobot, Canada.)
- Co-PI (PI: Amir Khajepour, Waterloo; 6 Co-PIs.)

Holistic Robot Farming — 2026 – 2028
- Mitacs Accelerate Grant ($560,000 CAD; applicant's share: 16.7%)
- Industry Partners: Haggerty AgRobotics and BHF Agrobot, Canada.)
- Co-PI (PI: Amir Khajepour, Waterloo; 6 Co-PIs.)

MATSEC: Materials-based Cybersecurity in Electronics — 2026 – 2028
- NCC ($2,000,000 CAD; applicant's share: 14.3%)
- Co-PI (PI: Sebastian Fischmeister, Waterloo.)

Data-driven traffic-aware driving style identification and simulation — 2025 – 2027
- Mitacs Accelerate Grant ($120,000 CAD; applicant's share: 100%)
- Industry Partner: Intact Insurance, Canada.)

Vulnerable road user detection in autonomous driving
— Research consulting grant ($27,040 CAD; applicant's share: 100%) 2025
— Industry Partner: Magna New Mobility, Canada.)

Scalable Flight Planning with Complex Airspace Restrictions and Fail-Safe Constraints
— NSERC Alliance Grant ($276,840 CAD; applicant's share: 50%) 2024 – 2026
— Co-PI (PI: Stephen L. Smith, Waterloo; Industry Partners: Navblue, Canada.)

Robot Learning from Demonstrations under Attack by Adversarial Experts
— CPI-RoboHub Seed Grant ($20,000 CAD; applicant's share: 50%) 2023 – 2024
— PI (Co-PI: Sirisha Rambhatla, Waterloo.)

Data-efficient Learning of Aircraft Dynamics for Simulated Flight Testing
— WISA Research-for-impact grant ($49,750 CAD; applicant's share: 50%) 2023 – 2024
— *WISA*
— Co-PI (PI: Vijay Ganesh, Waterloo; Co-PI: Kenneth Hui, NRC)

Real-time temporal logic-based motion planning for multi-agent systems in partially known environments.
— Discovery Grant ($145,000 CAD; applicant's share: 100%) 2022 – 2027
— *NSERC*

Real-time temporal logic-based motion planning for multi-agent systems in partially known environments.
— Discovery Launch Supplement ($12,500 CAD; applicant's share: 100%) 2022 – 2023
— *NSERC*

Scaled down robotic platforms for large scale autonomy.
— JELF ($80,000 CAD; applicant's share: 100%) 2023 – 2026
— *CFI*

Scaled down robotic platforms for large scale autonomy.
— ORF: Research Infrastructure ($80,000 CAD; applicant's share: 100%) 2023 – 2026
Start-up Grant ($175,000 CAD; applicant's share: 100%) 2021
— *University of Waterloo, Waterloo, ON*

Supervision

Current:

— Mingyue Deng, MASc, ECE Fall 2026 –
(co-advised with Prof. Roya Firoozi)
— Mark Chiu, MASc, ECE Fall 2026 –
(co-advised with Prof. Sebastian Fischmeister)
— Christian Schaible, PhD, ECE Fall 2025 –
(co-advised with Prof. Stephen L. Smith)
— Leroy D'Souza, PhD, ECE Fall 2023 –
(co-advised with Prof. Sebastian Fischmeister)
— Da Saem Lee, PhD, ECE Fall 2023 –
(co-advised with Prof. Sebastian Fischmeister)
— Melda Kiziltan, MASc (part-time), ECE Fall 2025 –
(co-advised with Prof. Stephen L. Smith)
— Matteo Golin, MASc, ECE Fall 2025 –
(co-advised with Prof. Stephen L. Smith)
— Lance Pharand, MASc, ECE Fall 2025 –
(co-advised with Prof. Stephen L. Smith)
— Zesheng (Ethan) Jia, M.Math, CS Fall 2025 –
(co-advised with Prof. Charles Clarke)

Former:

— Dr. Shamak Dutta, Postdoctoral Researcher, ECE — Fall 2024 – Winter 2026
(co-advised with Prof. Stephen L. Smith)
Next position: Research Scientist, Meta.

— Dr. Jie Wang, Postdoctoral Research Associate, ECE — Spring 2022 – Spring 2023
(co-advised with Prof. Sebastian Fischmeister)
Next position: Tenure-track Assistant Professor, Mechanical Engineering, University of Manitoba.

— Dr. Mohammad Pirani, Senior Research Associate, ECE — Fall 2021 – Winter 2022
(co-advised with Prof. Sebastian Fischmeister)
Next position: Tenure-track Assistant Professor, MME, University of Ottawa.

— Jiaming Zhong, PhD, MME — Spring 2022 – Winter 2025
(co-advised with Prof. Amir Khajepour)
Next position: Research Officer, National Research Council of Canada (NRC).

— Krishna Kalavadia, MASc, ECE — Fall 2024 – Spring 2026
(co-advised with Prof. Stephen L. Smith)
Next position: Software Engineer, Bloomberg.

— Praneeth Kolapalli, MASc, ECE — Spring 2024 – Winter 2026
(co-advised with Prof. Stephen L. Smith)
Next position: PhD Student, KTH (Sweden).

— Akash Karthikeyan, MASc, ECE — Fall 2023 – Spring 2025
Next position: PhD Student, Montreal Institute of Machine Learning (MILA).

— Arezoo Alipanah, MASc, ECE — Fall 2023 – Spring 2025
Next position: PhD Student, Computer Science, University of Waterloo.

— Avraiem Iskander, MASc, ECE — Fall 2023 – Spring 2025
(co-advised with Prof. Stephen L. Smith)
Next position: Software Engineer (Algorithms and Optimization), RideCo.

— Prajwal Thakur, MASc, ECE — Spring 2023 – Winter 2025
(co-advised with Prof. Amir Khajepour)
Next position: Autonomy Engineer, Addverb Robotics.

— Zhao-Feng (Jack) Dai, MASc, ECE — Fall 2022 – Fall 2024
(co-advised with Prof. Stephen L. Smith)
Next position: Software Engineer, Apple.

— Soroush Mortazavi Moghaddam, MASc, ECE — Fall 2022 – Spring 2024
(co-advised with Prof. Sebastian Fischmeister)
Next position: Data Scientist, Open Power.

— Leroy D'Souza, MASc, ECE — Spring 2022 – Spring 2023
(co-advised with Prof. Sebastian Fischmeister)
Next position: PhD Student, University of Waterloo.

Visiting researchers:

— Nathan Duncan, Rising Stars Visiting Undergraduate Researcher — Spring 2025
— Nikolaus Fischmeister, Visiting Researcher — Fall 2023
— Soham Lakhi, Visiting Undergraduate Researcher (UBC) — Fall 2022
— Harshil Naik, Visiting (remote) Undergraduate Researcher (IIT Bombay) — Spring 2022 – Fall 2022

Additional research supervision: 43 Undergraduate Research Assistants (1 ongoing); 5 Final-Year Design Project (capstone) teams; 9 MEng / graduate research-assistant projects.

Mentoring

Mentor: Soham Lakhi, visiting student, University of Waterloo, 2022
Mentor: Aayush Wadhwa, Graduate researcher, University of Waterloo, 2022
Next position: Software Engineer, Amazon, Canada.
Mentor: Harshil Kapadia, Graduate researcher, University of Waterloo, 2022
Next position: Software Engineer, Amazon, Canada.
Technical Expert: Girls in Engineering (GiE), University of California, Berkeley 2020
Senior Design Technical Advisor: Team UrbanDrone, ESE UPenn 2019
Senior Design Technical Advisor: Team EagleEye, ESE UPenn 2018
— Winners of the 2018 Frederick Ketterer Memorial Award
— Winners of the 2018 Federal Aviation Administration (FAA) RAISE Award

Selected Service

Departmental and Faculty committees:
— Departmental Advisory Committee on Appointments, Controls & Decision tenure-track search, ECE — Member 2023 – 2024
— Advisory Committee on Engineering Scholarships, Faculty of Engineering — ECE Representative 2023–
— Engineering Faculty Council, Faculty of Engineering — Member 2022 – 2023
Other university service:
— Co-organizer (with Prof. Pascal Poupart), Waterloo.AI Institute Seminar Series 2022 – 2025
— Faculty Marketing & Communications: multi-robot demonstration and video promoting the Faculty of Engineering's research 2025
— Research and Strategic Plan consultation with early-career researchers (ECRs), U. Waterloo 2025
— Consultant, Capstone Projects, ECE, U. Waterloo 2022–
External service:
— Member, Advisory Board, Banff International Research Station for Mathematical Innovation and Discovery (BIRS) 2026–
Graduate examination committees:
— 8 PhD Defense committees
— 20 PhD comprehensive and background examination committees
— 6 PhD seminar committees
— 16 Master's (MASc) thesis-reader committees
— 3 ECE 499 undergraduate-report reader

Reviewing Activities

Journal reviewer
IEEE Transactions on Automatic Control (TAC)
IEEE Transactions on Robotics (T-RO)
IEEE Robotics and Automation Letters (RA-L)
IEEE Transactions on Intelligent Transportation Systems
IEEE Transactions on Control of Network Systems (TCNS)
IEEE Control Systems Letters (L-CSS)
IEEE Embedded Systems Letters
Autonomous Robots (Springer)
Journal of Intelligent & Robotic Systems
IET Intelligent Transport Systems
Journal of Artificial Intelligence Research (JAIR)
Journal of Automated Software Engineering (AUSE)
ACM Transactions on Embedded Computing Systems
Proceedings of the IEEE
Chemical Product and Process Modeling (CPPM)

Book reviewer, MIT Press

Conference & workshop reviewer

Conference on Robot Learning (CoRL)	2026
IEEE International Conference on Intelligent Transportation Systems (ITSC)	2022 – 2026
AIA Workshop at IJCAI	2025
European Control Conference (ECC)	2025
American Control Conference (ACC)	2014, 2018, 2020 – 2026
IEEE Conference on Control Technology and Applications (CCTA)	2024
Conference on Learning for Dynamics and Control (L4DC)	2020, 2021, 2023
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	2023
Indian Control Conference (ICC)	2015 – 2017, 2022, 2023
IEEE Conference on Decision and Control (CDC)	2016, 2019, 2022
IEEE International Conference on Robotics and Automation (ICRA)	2021, 2023
International Conference on Communications, Computation, Networks and Technologies (INNOV)	2019
Design Automation Conference (DAC)	2018
ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)	2013 – 2018
International Conference on Embedded Software (EMSOFT)	2013, 2015, 2018
Symposium on Reliable Distributed Systems (SRDS)	2015
ACM Conf. on Languages, Compilers, Tools & Theory for Embedded Systems (LCTES)	2015
ACM International Conference on Future Energy Systems (e-Energy)	2015
International Conference on Information Processing in Sensor Networks (IPSN)	2015
European Conference on Wireless Sensor Networks (EWSN)	2015
ACM International Conference on High Confidence Networked Systems (HiCoNS)	2014
IEEE Real-Time Systems Symposium (RTSS)	2012
IEEE International Conference on Sensing, Communication and Networking (SECON)	2012

Grant & professional-licensure reviewer Canada Foundation for Innovation (CFI) — John R.

Evans Leaders Fund (JELF) proposal	2025
NSERC Discovery Grant application	2024 – 2025
NSERC Alliance Grant application	2024
Professional Engineers Ontario (PEO) P.Eng. competency review	2024

Conference Services

Editorial

Associate Editor: IEEE Conference on Control Technology and Applications (CCTA)	2024–
Editorial Board Member: IEEE Control Systems Society (CSS) Technical Conferences	2023–
Associate Editor: Indian Control Conference (ICC)	2023 – 2024
Editorial Board Member: Journal of Software Engineering for Autonomous Systems	2023 – 2024
Associate Editor: The 2024 American Control Conference (ACC)	2023

Conference and workshop organization

Program Chair: Conference on Robots and Vision (CRV)	2027
Program Committee Member: Canadian AI Conference (CAIAC)	2025–
Session Chair: Canadian AI Conference (CAIAC)	2025
Program Committee Member: AAAI Conference on Artificial Intelligence	2025
Program Committee Member: 3rd International Conference on Transportation Infrastructure Projects: Conception to Execution (TIPCE)	2025
Program Committee Member: IEEE/ACM International Conference on Cyber-Physical Systems (ICCPS)	2024 – 2025
Program Committee Member: IEEE Conf. on Embedded and Real-Time Computing Systems and Appli-	

cations (RTCSA) 2021 – 2024

Publicity Co-chair: IEEE/ACM International Conference on Cyber-Physical Systems (ICCPS) 2023

Session Chair (Autonomous Systems): IEEE Conference on Control Technology and Applications (CCTA) 2022

Program Committee Member: 2nd Workshop on Opportunities & Challenges with Autonomous Racing (OCAR), at ICRA 2022

Workshop Co-chair & Organizer: Workshop on Intelligent Transportation Systems (ITS), at COMSNETS 2021 – 2022

Program Committee Member: 9th International Conference on Communications, Computation, Networks and Technologies (INNOV) 2020

Program Committee Member: 4th Workshop on the Design and Analysis of Robust Systems (DARS), at CAV 2019

Technical Committee Member: 8th International Conference on Communications, Computation, Networks and Technologies (INNOV) 2019

Selected Talks

" Diffusion-based generative models for autonomous system behavioral planning"
— Brock University Data Science and Artificial Intelligence Seminar. August 2025

"Signal Temporal Logic-based Motion planning for Multi-Robot Systems with Complex Objectives"
— International Workshop on Design Automation for CPS and IoT (DESTION). May 2025

"Adaptformer: Sequence Models as Adaptive Iterative Planners"
— The Canadian Applied and Industrial Mathematics Society (CAIMS), Annual Meeting. June 2024

"Motion planning for Multi-Agent Systems with Signal Temporal Logic objectives"
— Meeting on Systems and Control Theory (MCST). May 2023
— UC Santa Cruz Cyber-Physical Systems Research Center (CPSRC). March 2023

"Distributed planning for drone fleets with Temporal Logic Objectives"
— IEEE CCTA August 2022
— Intel-UPenn annual visit, Philadelphia, USA May 2019

"Fly-by-Logic: Control of Multi-rotor drone fleets using Temporal Logic Objectives" (and related topics)
— Indian Institute of Technology Roorkee, India August 2021
— Aalto University, Finland November 2021
— Qualcomm Research - Autonomous Driving R&D, USA June 2020
— NSF VeHICaL Annual Meeting, UC Berkeley, USA October 2019
— Stanford Autonomous Systems Lab (ASL), USA July 2019
— Nokia-Bell Labs, Murray Hill, USA Jun 2019
— NASA Formal Methods Symposium, Houston, USA May 2019
— Monthly WAS Intel Science and Technology Center (ISTC) seminar¡ USA Aug 2018
— Amazon Robotics, Boston, USA Jul 2018
— Mathworks Research, Boston, USA Jul 2018
— ICCPS at CPS Week, Porto, Portugal Apr 2018
— UPenn GRASP/ PRECISE Industry Symposium (Poster and Demonstration), Philadelphia, USA Feb 2018

"Smooth Operator: Control using the Smooth Robustness of Metric Temporal Logic"
— IEEE CCTA, Big Island, Hawaii, USA Aug 2017
— SRC Techcon (Poster), Austin, USA Sep 2017

"Robust Model Predictive Control for Non-Linear Systems with Input and State Constraints via Feedback Linearization"
— CDC, Las Vegas, USA Dec 2016

"Co-design of Anytime Computation and Robust Control"

— UPenn ESE PhD Colloquium, Philadelphia, USA — Mar 2016

— UPenn PRECISE Industry Day (Poster), Philadelphia, USA — Feb 2016

— RTSS, San Antonio, USA — Dec 2015

Teaching Experience

Instructor

— MTE 484: Digital Control Applications, U. Waterloo — Fall 2026

— ECE 380: Analog Control Systems, U. Waterloo — Winter 2023, 2024, Spring 2024

— ECE 481: Digital Control Systems, U. Waterloo — Spring 2022, 2023, 2024

— SE 380: Feedback Control Systems, U. Waterloo — Fall 2025

— ECE 488: Multivariable Control Systems, U. Waterloo — Winter 2026

— ECE 780: Model Predictive Control, U. Waterloo — Winter 2022, Spring 2023

Guest Lecturer

— ESE 350: Embedded Systems, U. Penn, Philadelphia, USA — Apr 2019

— ESE 519: Embedded Systems, U. Penn, Philadelphia, USA — Nov 2018

— CSCI 699: Formal methods for Cyber-Physical Systems, USC, USA — Oct 2018

Teaching Assistant

— ESE 406/505: Control Theory, U. Penn, Philadelphia, USA — Fall 2014

— ESE 406: Control Theory, U. Penn, Philadelphia, USA — Fall 2013